

Data Literacy for Political Science

POLI-LIT

Outline

- What is POLI-LIT? Why data literacy?
- Political Science Research Methods
- Making Statistical Inference
- Conceptualization to Measurement - V Dem EDI
- POLI LIT Unit 1 Preview

What is POLI-LIT?

Political Data Literacy module

Edwina Chen and Insu Hwang

Political Science

Edwina Chen is a Ph.D. candidate in Political Science. Originally from Taiwan, she earned her bachelor's in political science from National Cheng Kung University (TW) and a master's in politics from the University of Essex (UK). Her research interests include LGBTQ+ politics, queer theory, authoritarianism, democratization, and East Asian politics. Her current work examines global patterns in same-sex marriage legalization and how voters perceive LGBTQ+ political candidates. Outside of her academic work, Chen loves traveling and learning from the cultures and stories she encounters along the way. Professional website: edwinachen.com



Edwina Chen

Insu Hwang is a Ph.D. student in Political Science at USC. He studies varied topics in International Relations and Comparative Politics. He earned his B.A. in International Studies and an M.A. in International Studies at Pukyung National University in Busan, Republic of Korea. His research interest includes the re-balancing of Global Power, particularly focusing on the concept of 'Middle Power' amidst the rise of Indo-Pacific region, as well as Asian security and the role of International Institutions.



Insu Hwang

Project: R and Data Literacy Module for Political Science Undergraduates

This project will develop a self-paced, web-based instructional module for POLI 101: Introduction to Global Politics. The resource aims to strengthen data literacy by introducing students to political datasets and the basics of R programming, while also teaching usage of artificial intelligence tools such as ChatGPT for learning support. By integrating real-world datasets, the module will help students interpret data structures, understand descriptive statistics, and create simple visualizations. Deliverables will include a structured module (3–4 units), an instructor manual with guidance on course integration, assessments, and a curated resource hub. Although designed for POLI 101, the module will be adaptable for asynchronous or hybrid use in other political science or social science courses.

What is POLI-LIT?

Political Data Literacy module

Designed to provide you with:

- **Data skills** — Learn how to read and interpret datasets and codebooks
- **R skills** — Use R to organize, summarize, and visualize political data
- **AI literacy** — Use tools like ChatGPT to assist (not replace!) your analysis
- **Critical thinking** — Question how data is constructed and used in political arguments

Why Data Literacy Matters in Politics

Many political claims rest on data:

“Crime is rising in urban areas.” “Voter turnout declined last year.”

“Democracy is eroding globally.”

But

Who collected the data?

How was *crime* or *democracy* defined?

What choices were made in measurement?

Why Data Literacy Matters in Politics

Many political claims rest on data; without data literacy, you consume what is fed to you.

POLI-LIT gives you tools to answer those questions and interpret claims.

We want you not just to consume political data, but to interrogate it.

Political Science Research Methods

How We Investigate

PS
RMI

Political Science
Research and Methods

THE JOURNAL OF THE EUROPEAN POLITICAL SCIENCE ASSOCIATION

ISSN 2049-8470

Types of Research

We asks questions in three broad ways:

Descriptive:

What is happening? → Describe facts, trends, frequencies

“What percent of eligible citizens voted?”

Types of Research

We asks questions in three broad ways:

Predictive:

What will happen? → Forecast outcomes

Example: “Which party will win the next election?”

Types of Research

We asks questions in three broad ways:

Explanatory:

Why / how did that happen? → Investigate causes, mechanisms

Example: “Why do some democracies collapse?”

Types of Research

What type of research is this?

Descriptive: What is happening? → Describe facts, trends, frequencies

Predictive: What will happen? → Forecast outcomes

Explanatory: Why / how did that happen? → Investigate causes, mechanisms

“Will voter turnout increase if polling stations open on weekends?”

Types of Research

What type of research is this?

Descriptive: What is happening? → Describe facts, trends, frequencies

Predictive: What will happen? → Forecast outcomes

Explanatory: Why / how did that happen? → Investigate causes, mechanisms

“What percent of countries held national elections during the COVID-19 pandemic?”

Types of Research

What type of research is this?

Descriptive: What is happening? → Describe facts, trends, frequencies

Predictive: What will happen? → Forecast outcomes

Explanatory: Why / how did that happen? → Investigate causes, mechanisms

“Why did support for democracy decline among young voters after 2020?”

The line between qualitative and quantitative research is increasingly blurred — many projects use both.

But if we ask:

What type of data are we using?

and

What kind of analysis are we doing?

— we can still understand the key differences and choose the right approach for our question.

Types of data

Identify patterns, trends, and relationships

Surveys (e.g., Public opinion polls)

Election Data (e.g., Voter turnout, party vote share)

Index Scores (e.g., Democracy Index, Corruption Perceptions Index)

Government & Economic Data (e.g., GDP, military spending, trade policies)

Types of data

Identify patterns, trends, and relationships

Surveys, Election Data, Index Scores, Government & Economic Data...

Also

Interviews & Focus Groups (e.g., In-depth conversations with voters about why they support a candidate)

Field Notes / Ethnographic Observation (e.g., Observing a town hall meeting, a protest)

Types of data

Archival Records ? Policy Documents & Speeches? Media & Content Analysis?

Normalization of Censorship: Evidence from China

Tony Zirui Yang

≡ Abstract

≡ Full Text

PDF

Supplemental Material



Abstract

Authoritarian censorship is traditionally understood as a repressive tool and will cause public backlash against the regime. However, surveys worldwide consistently find that citizens in authoritarian regimes are apathetic toward or even supportive of government censorship. To explain this puzzle, I theorize that citizens are subject to a normalization process. Specifically, individuals become desensitized to censorship activities when the range of censored content expands beyond politically threatening topics, like government criticism and collective action, to other less political issues. Using 28 million censored articles and two original survey experiments in China, I show that (1) a majority of censored articles are indeed unrelated to politically threatening topics, and (2) respondents exposed to the censorship of both political and nonpolitical content report higher support for censorship and the regime than those exposed only to political censorship. These findings highlight how normalizing repressive apparatuses like censorship contributes to authoritarian control and survival.

Although documents and texts are traditionally used in qualitative research, they can also be analyzed quantitatively using methods like text analysis or content coding.

Although documents and texts are traditionally used in qualitative research, they can also be analyzed quantitatively using methods like text analysis or content coding.

For example, we can:

Count how often certain words or phrases appear in political speeches or news articles

Measure sentiment (positive/negative tone) in social media posts

Track frequency of themes across hundreds of documents over time

This turns **words into data**, allowing us to detect patterns, compare across cases, and run statistical tests — all using quantitative tools.

Types of Analysis

Qualitative : Discourse analysis, Case studies/ process tracing, Narrative analysis...

Quantitative : Descriptive statistics, Correlation analysis , Regression analysis...

Many political science projects combine methods: Mixed Methods

The method you choose should align with your **research question**.

Research Question	Data Type	Analysis Type
1. How often do first-generation college students attend office hours?	Survey data on office hour attendance & demographics	Descriptive statistics (frequencies, means) – e.g. t-test between groups
2. Why did youth voter turnout rise in the 2020 U.S. election?	Interview transcripts with young voters, campaign organizers	Thematic coding to identify motivations, narratives, influences
3. How do local politicians talk about immigration in media, and how does it affect voters' attitudes?	Speeches, social media posts (text); pre/post survey on voter attitudes	Text analysis (e.g. sentiment, topic modeling) + regression to test media exposure effect

Which part is qualitative, which is quantitative?

Strengths & Limitations

Aspect	Qualitative	
Strengths	Deep understanding, context, meaning Reveals mechanisms & nuance	
Limitations	Hard to generalize, time/resource intensive Small sample size, subjective	
Useful for	Case-specific questions	

Each method has trade-offs. Choose based on your research goals.

Strengths & Limitations

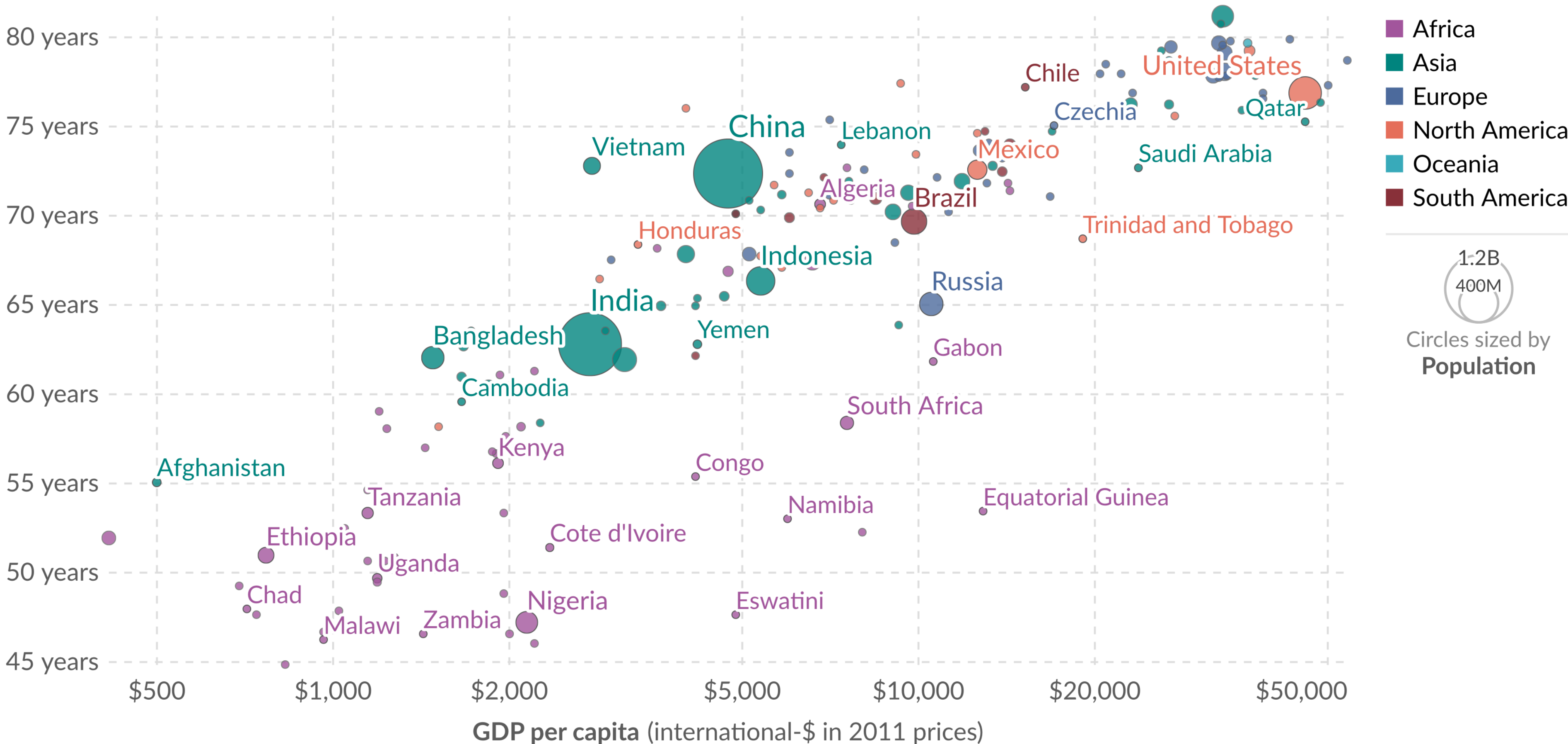
Aspect	Qualitative	Quantitative
Strengths	Deep understanding, context, meaning	Breadth, generalizability, statistical testing
	Reveals mechanisms & nuance	Can identify patterns across many cases
Limitations	Hard to generalize, time/resource intensive	May miss context or meaning
	Small sample size, subjective	Dependent on good data quality
Useful for	Case-specific questions	Broad comparisons

Each method has trade-offs. Choose based on your research goals.

Life expectancy vs. GDP per capita, 2000

The period life expectancy¹ at birth, in a given year. GDP per capita is adjusted for inflation and differences in living costs between countries.

Life expectancy at birth



Higher GDP = Longer life?

What Is Statistical Inference?

In political science, we often want to ask:

“Is there a meaningful relationship between two variables?”

Statistical inference:

The process of analyzing data generated by a sample, but then used to determine some characteristic of the larger population.

What Is Statistical Inference?

“Is there a meaningful relationship between two variables?”

Statistical inference helps us test this using sample data.

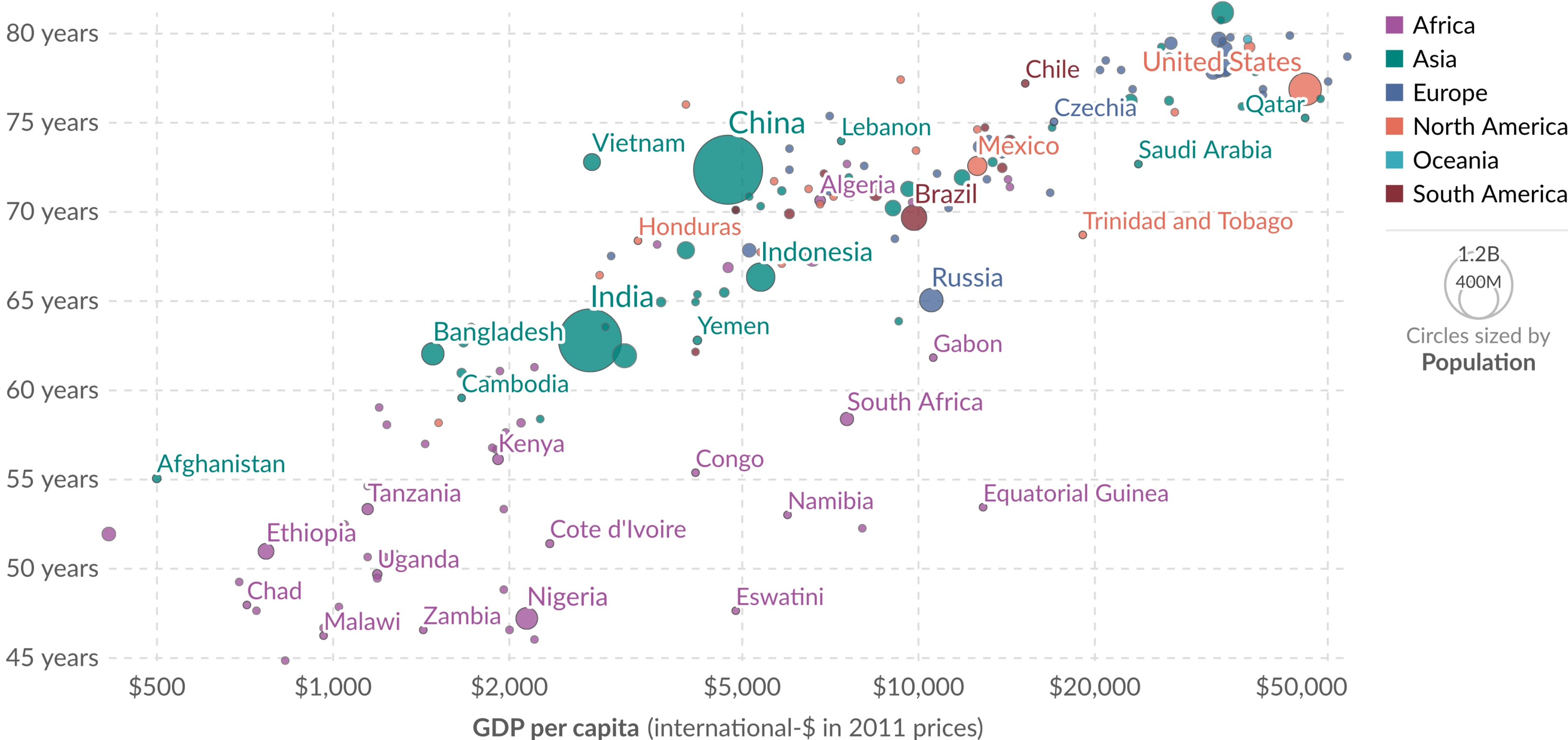
- Independent Variable (IV): the “cause” (what we change or compare)
- Dependent Variable (DV): the “effect” (what we observe)

➤ We check if the IV and DV are related using statistical tests.

Life expectancy vs. GDP per capita, 2000

The period life expectancy¹ at birth, in a given year. GDP per capita is adjusted for inflation and differences in living costs between countries.

Life expectancy at birth



IV (cause)? DV(effect)?

Key concepts:

Independent Variable (IV): the “cause”

Dependent Variable (DV): the “effect”

Correlation = two variables move together

Causation = one variable *produces* change in another

➤ Just because two things are correlated doesn't mean one caused the other!



Example: T-test

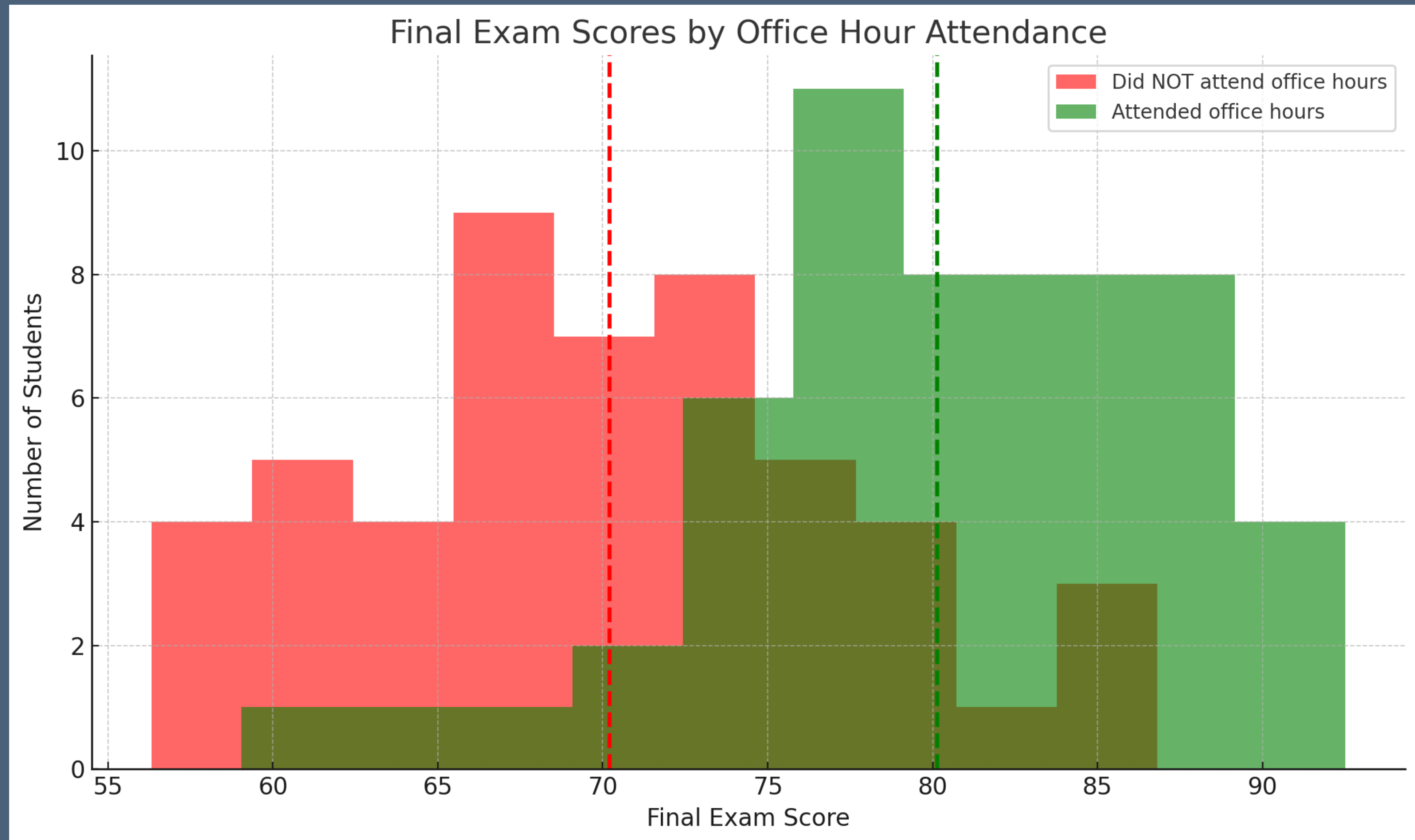
Do students who attend office hours score higher on the final exam?

IV: Attending office hours (Yes/No)

DV: Final exam score

Method: Compare the mean (exam score) of two groups?

Do students who attend office hours score higher on the final exam?



Example: T-test

Do students who attend office hours score higher on the final exam?

IV: Attending office hours (Yes/No)

DV: Final exam score

Method: Compare the mean (exam score) of two groups?

A t-test is a statistical test used to compare means between two groups. It helps determine whether the difference between the groups is statistically significant or just due to chance.

Example: T-test

Do students who attend office hours score higher on the final exam?

IV: Attending office hours (Yes/No)

DV: Final exam score

Method: T test. Compare the mean (exam scores) of two groups.

Group	N	Mean Score	Std. Dev.
Did NOT attend office hours	50	72.1	7.9
Attended office hours	50	80.3	8.1

Do students who attend office hours score higher on the final exam?

IV: Attending office hours (Yes/No)

DV: Final exam score

P- Value:

A number (between 0 and 1) that tells us how likely our results are due to random chance

Common threshold: $p < 0.05$ = less than 5% chance it happened randomly

Statistic	Value
t-statistic	5.18
p-value	0.00001
df (degrees of freedom)	98
Interpretation	$p < 0.05 \rightarrow$ statistically significant

Do students who attend office hours score higher on the final exam?

Common threshold: $p < 0.05$ = less than 5% chance it happened randomly

Smaller p-value = more evidence that there's a real relationship

➤ If $p < 0.05 \rightarrow$ we might say office hours *make a difference*

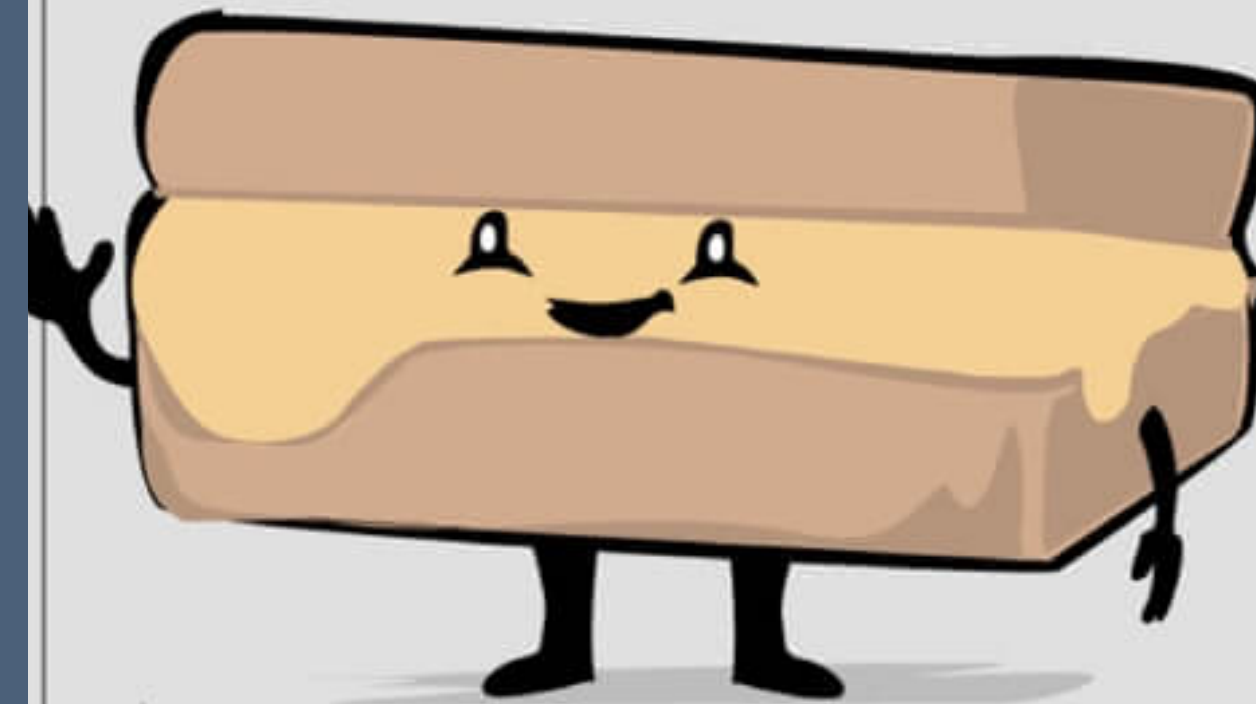
➤ If $p > 0.05 \rightarrow$ difference could just be random

Statistic	Value
t-statistic	5.18
p-value	0.000001
df (degrees of freedom)	98
Interpretation	$p < 0.05 \rightarrow$ statistically significant difference

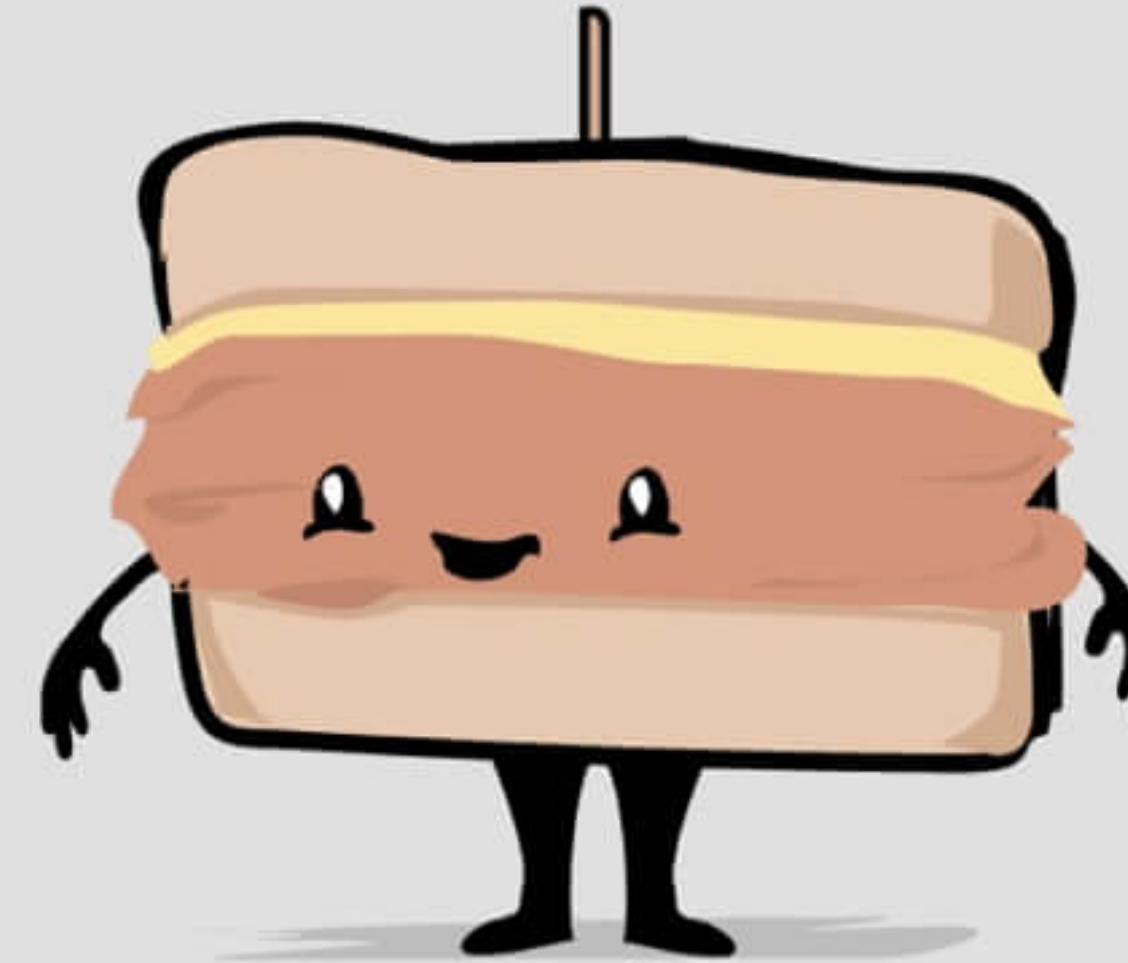
Conceptualization to Measurement

V Dem EDI

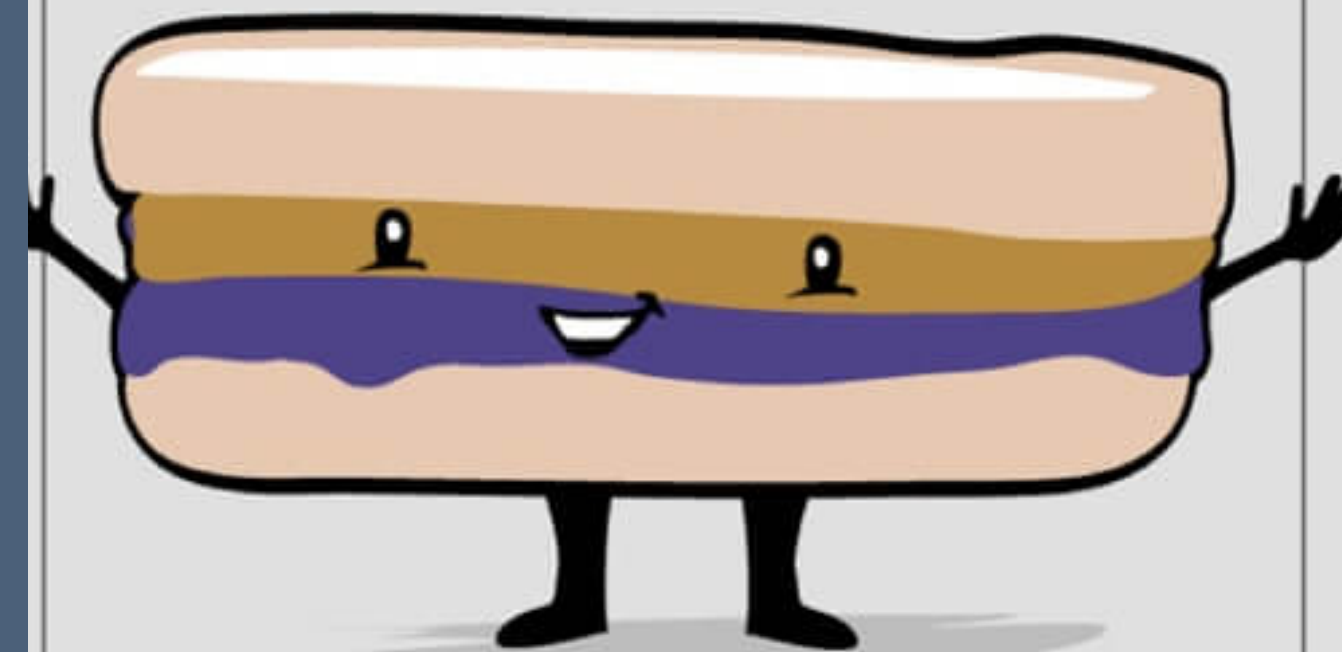
Classic Sandwiches



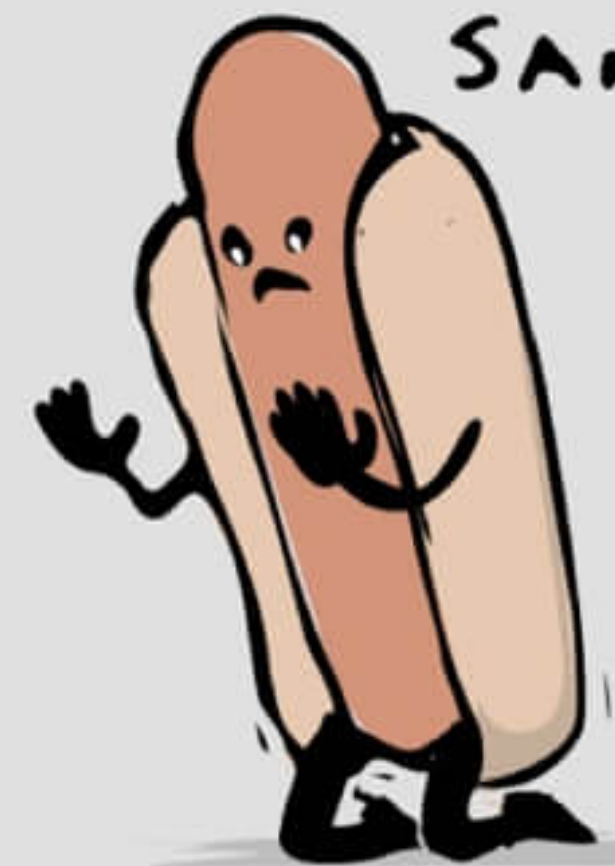
Grilled Cheese



Reuben



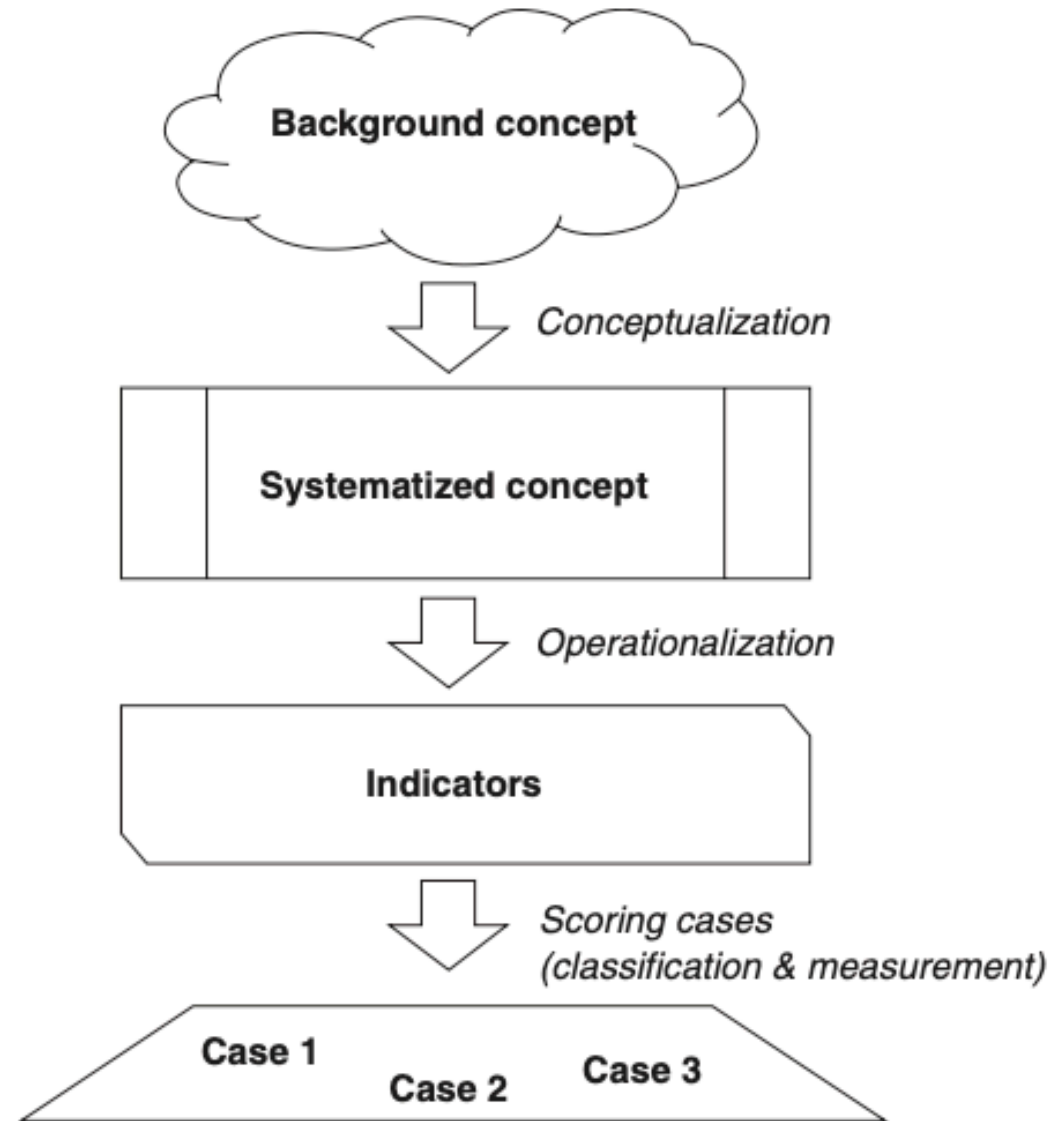
PBJ



Hot Dog

AM I A
SANDWICH

- Democracy



From Concept → Measurement (Process)

To study an abstract idea (like democracy, legitimacy, representation), we follow this pipeline:

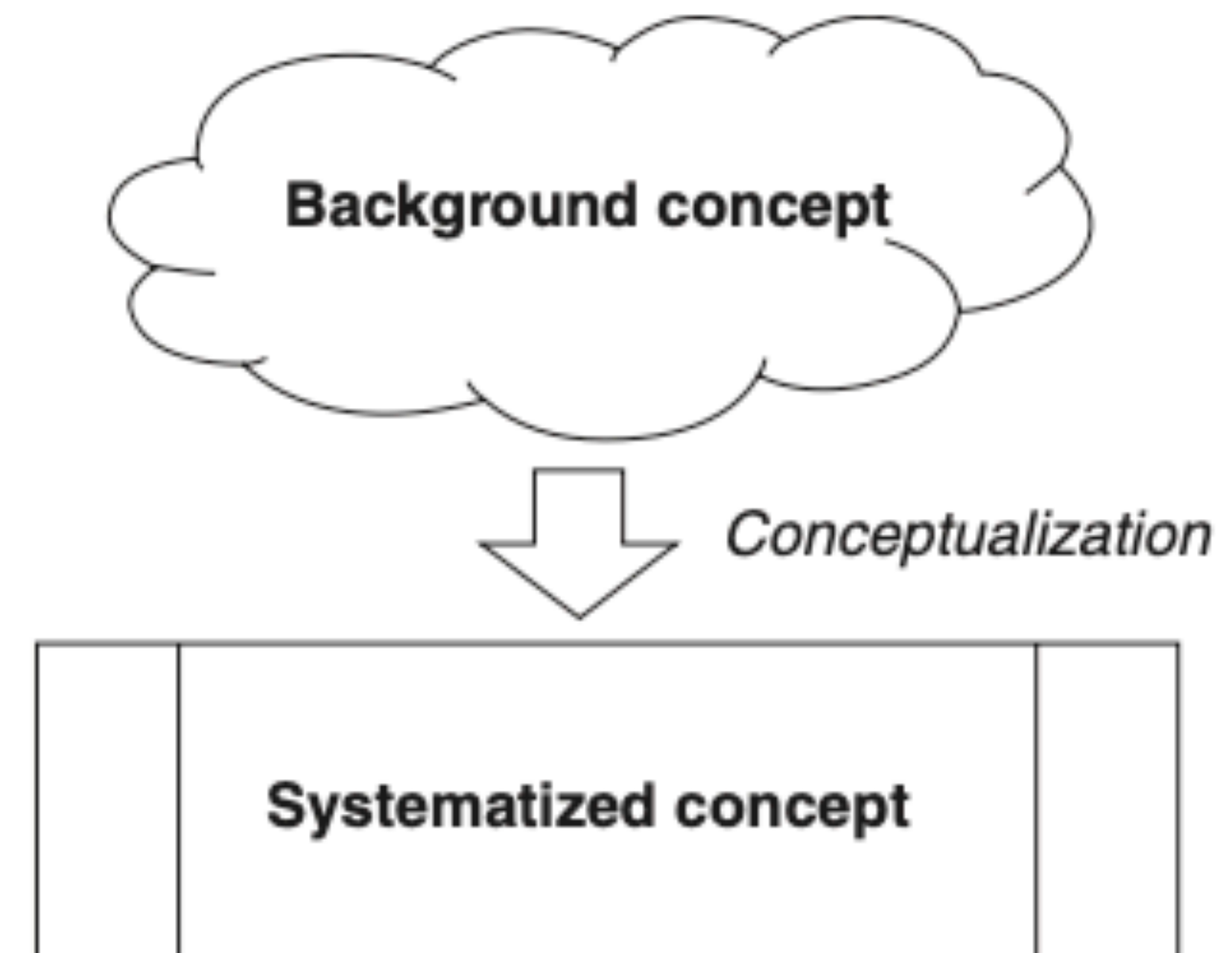
1. Conceptualization: What does the idea mean?
2. Indicator(s): What observable things will represent that concept?
3. Measurement / Coding: How do we assign values (scores, categories) to those indicators?

Conceptualization

A concept is a mental or theoretical notion (e.g., democracy, freedom, representation).

Conceptualization is the process of defining exactly what we mean by that concept in our study.

Figure 4.2 *Conceptualization and operationalization*



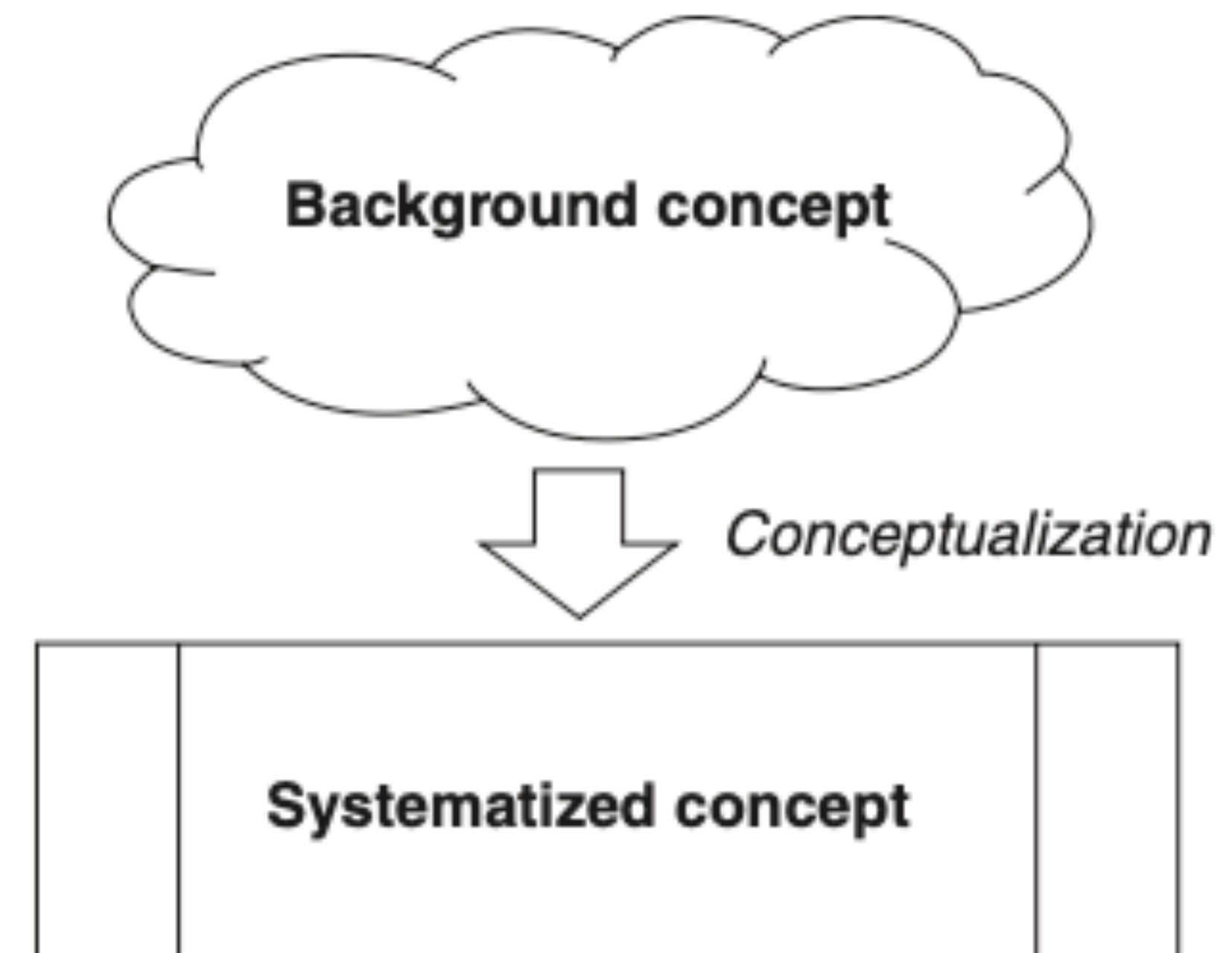
Conceptualization

A concept is a mental or theoretical notion (e.g., democracy, freedom, representation).

Conceptualization is the process of defining exactly what we mean by that concept in our study.

The process of creating mental categories that capture the meaning of objects, events, or ideas.

Figure 4.2 *Conceptualization and operationalization*

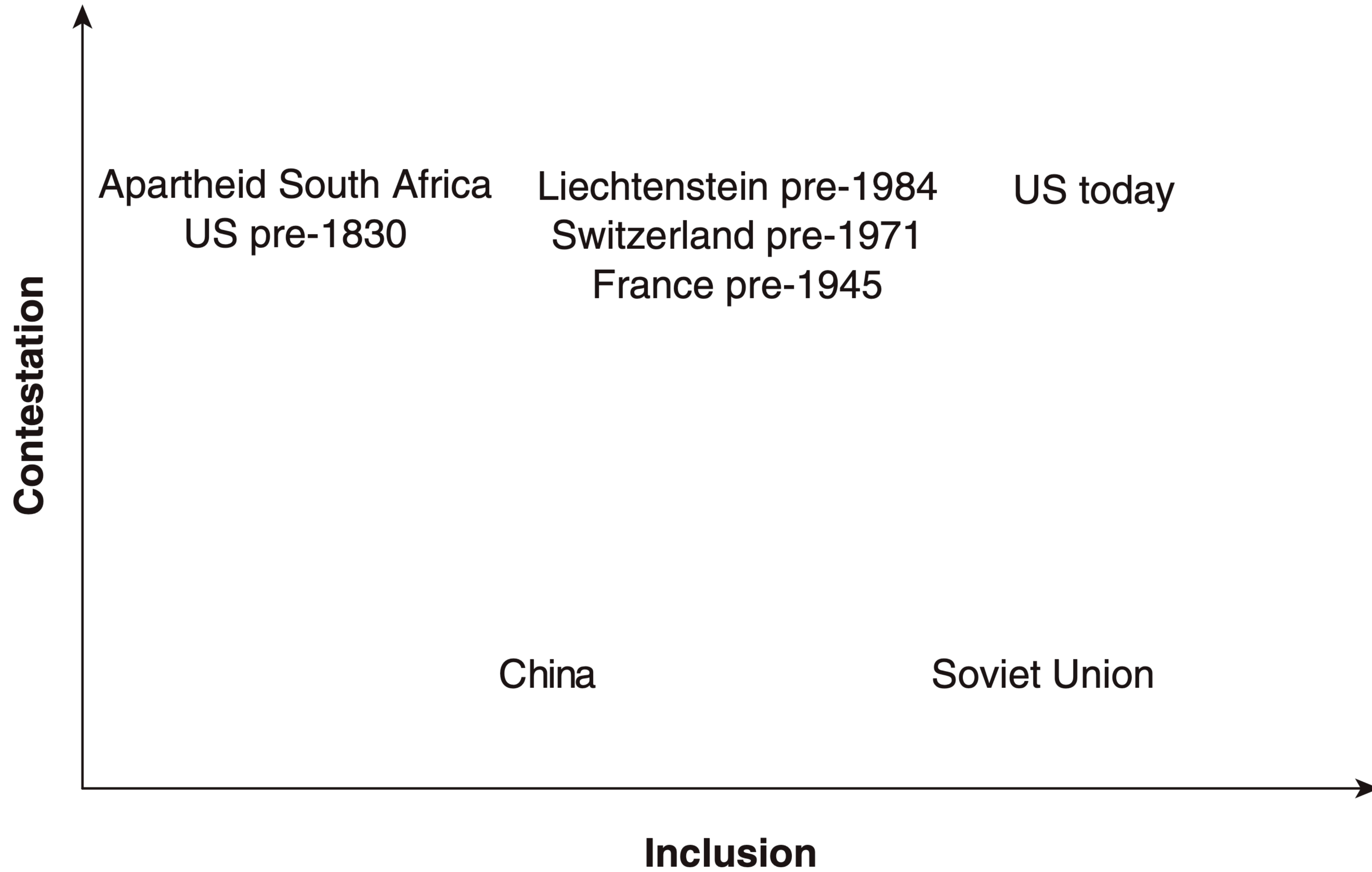


Conceptualization: What *does* democracy mean?

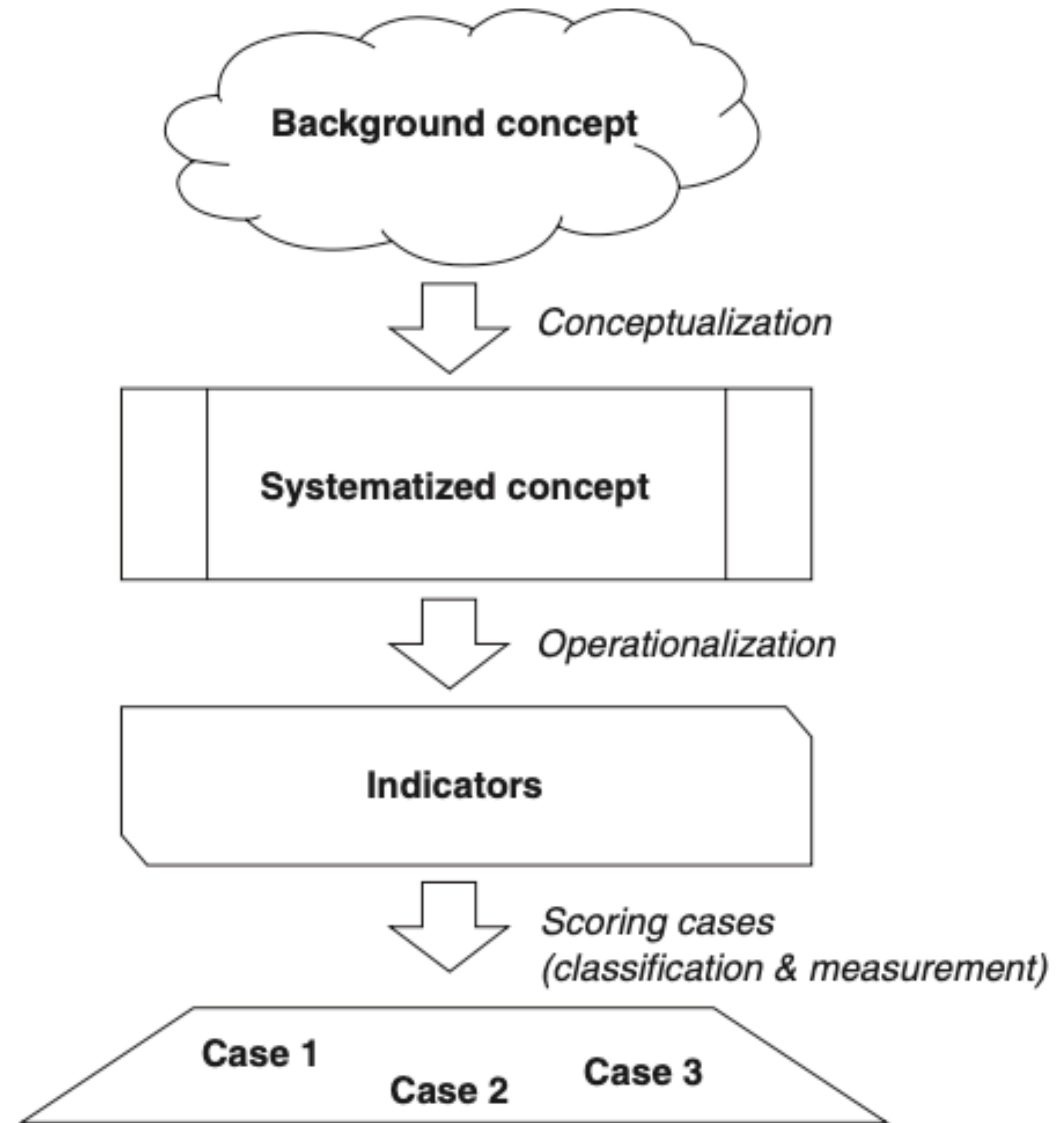
Robert Dahl proposed a minimalist view of democracy.

Two dimensions:

1. Contestation captures the extent to which citizens are free to organize themselves into competing blocs in order to press for the policies and outcomes they desire.
2. Inclusion has to do with who gets to participate in the democratic process.



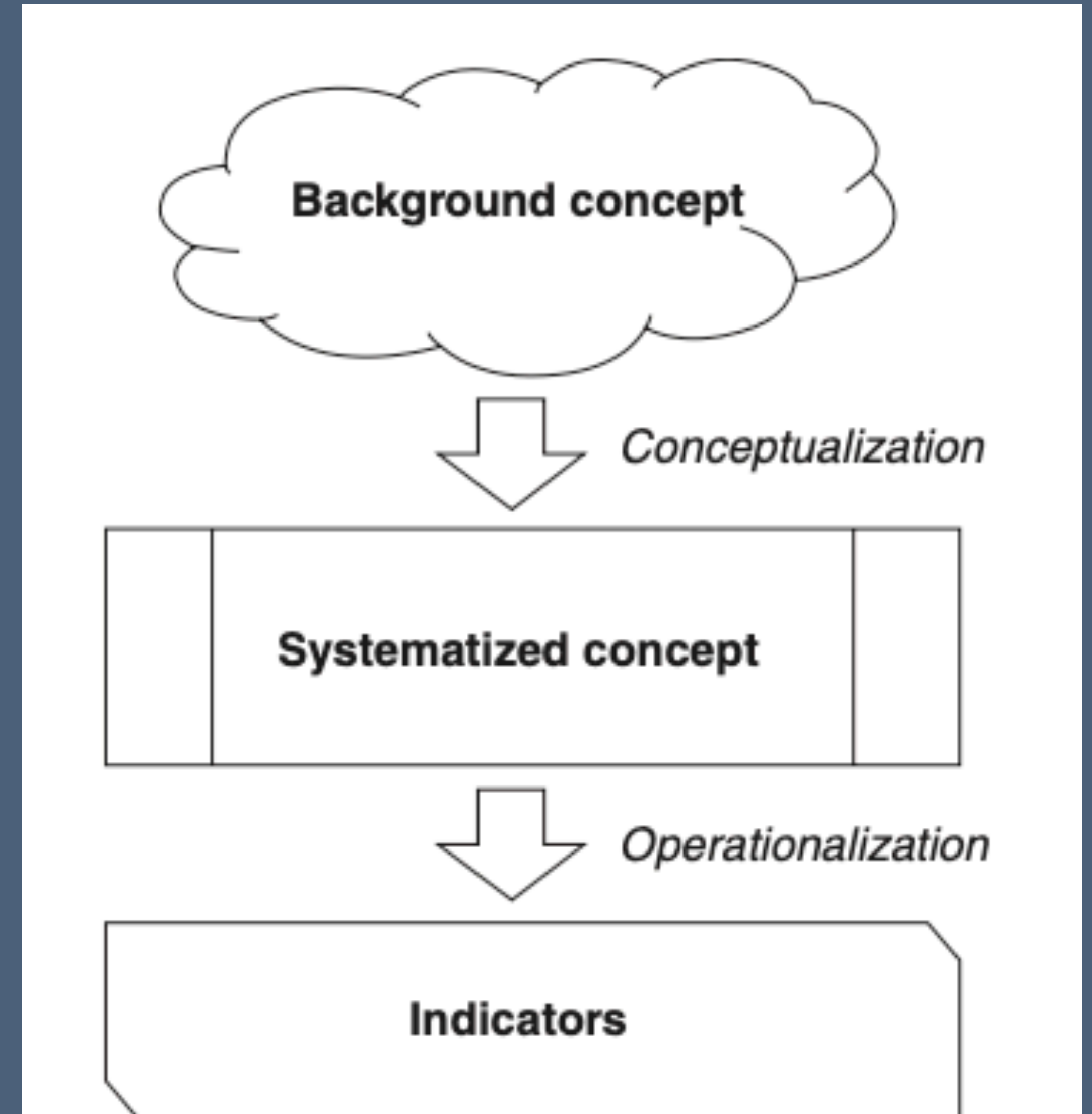
- Democracy
- Contestation and Participation/Inclusion



Operationalization

— Choosing Indicators

Indicators are observable phenomena that stand in for the abstract concept.

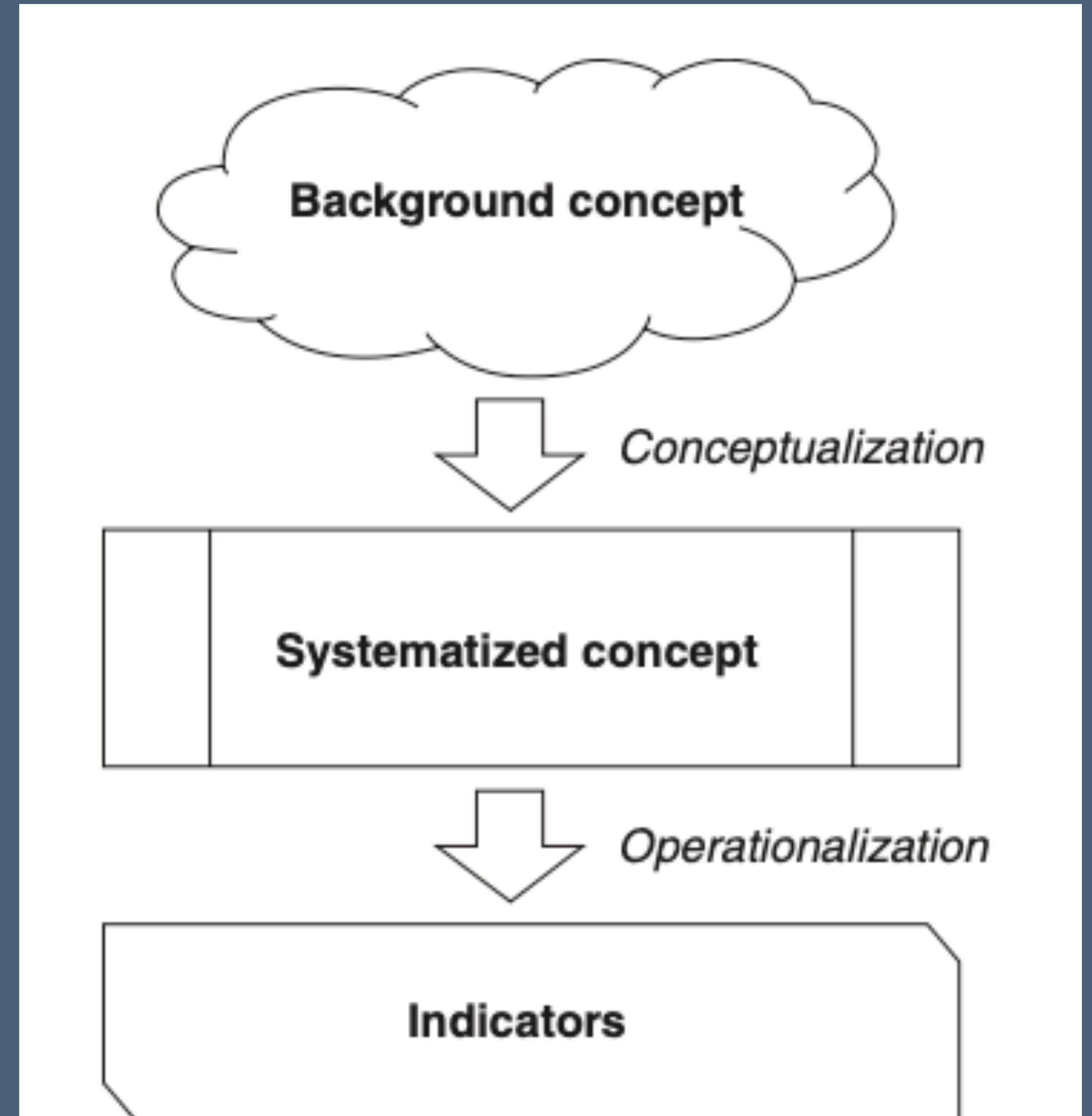


Operationalization

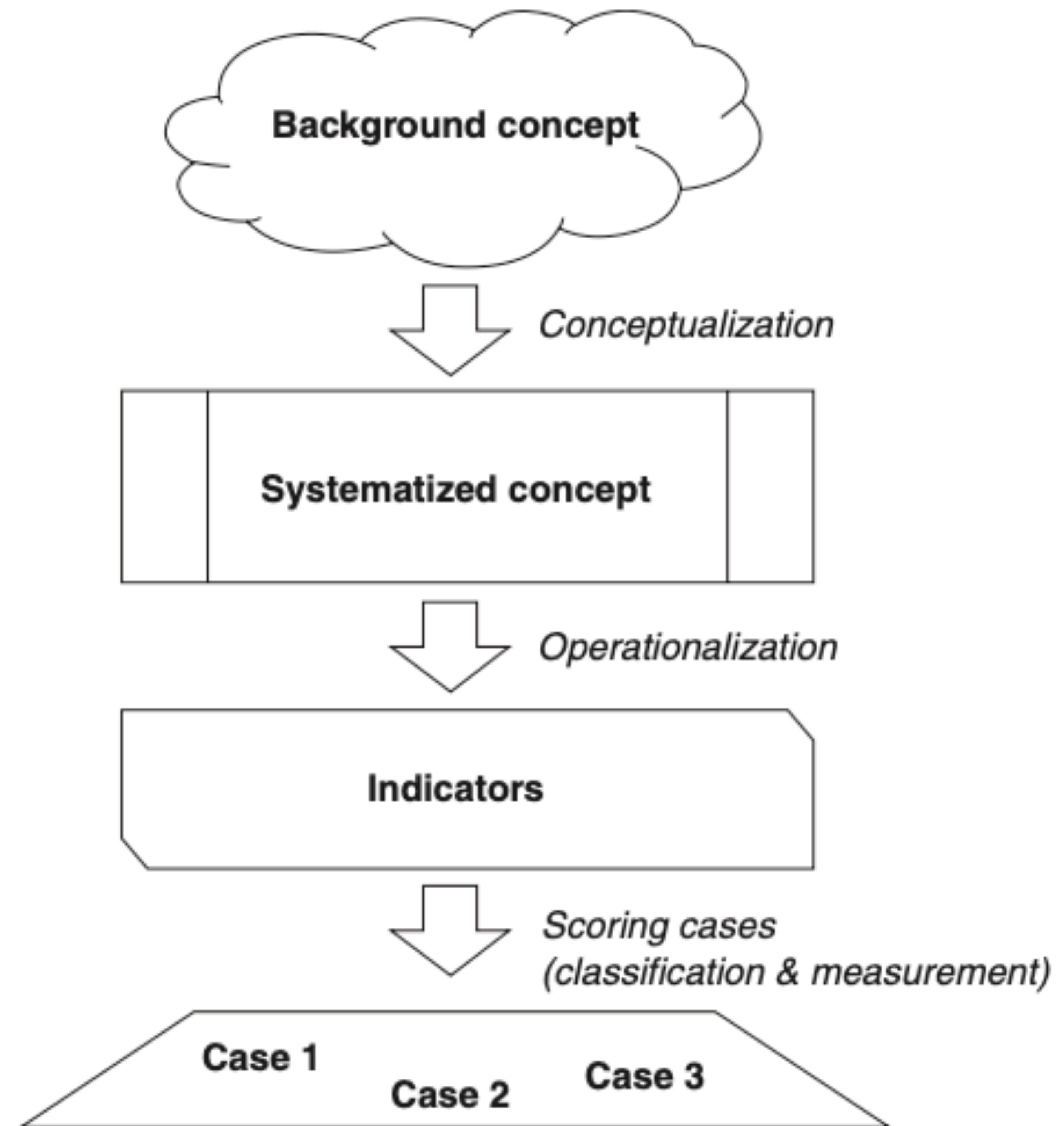
— Choosing Indicators

For Dhal's concept of democracy, possible indicators include:

- “Were elections free & fair?”
- “Freedom of speech / media”
- “Right of association (political parties, civic groups)”
- “Extent of suffrage / inclusion”

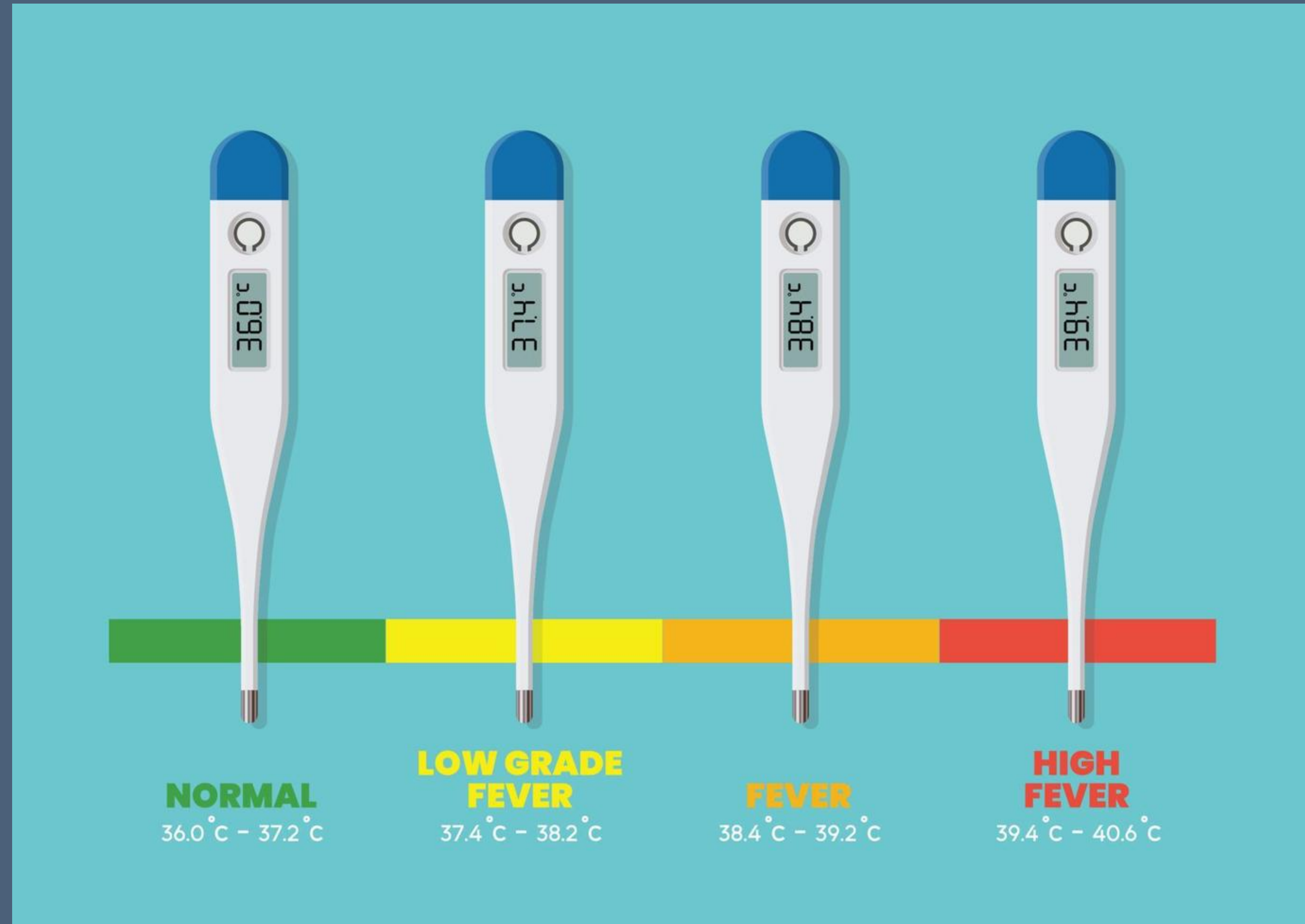


- Democracy
- Contestation and Participation/Inclusion
- Regular election? Voting Right? Effective party counts?



Varieties of Democracy	Narrow and broader: electoral, liberal, participatory, deliberative, or egalitarian democracy
Regimes of the World	Narrow: electoral or liberal democracy
Lexical Index	Narrow: electoral (or liberal) democracy
Boix-Miller-Rosato	Narrow: electoral democracy
Polity	Narrow: electoral and liberal democracy
Freedom House	Narrow: electoral or liberal democracy
Bertelsmann Transformation Index	Broad: electoral, liberal, participatory, deliberative, and effective democracy
Economist Intelligence Unit	Broad: electoral, liberal, participatory, deliberative, and effective democracy

Measurement / Coding

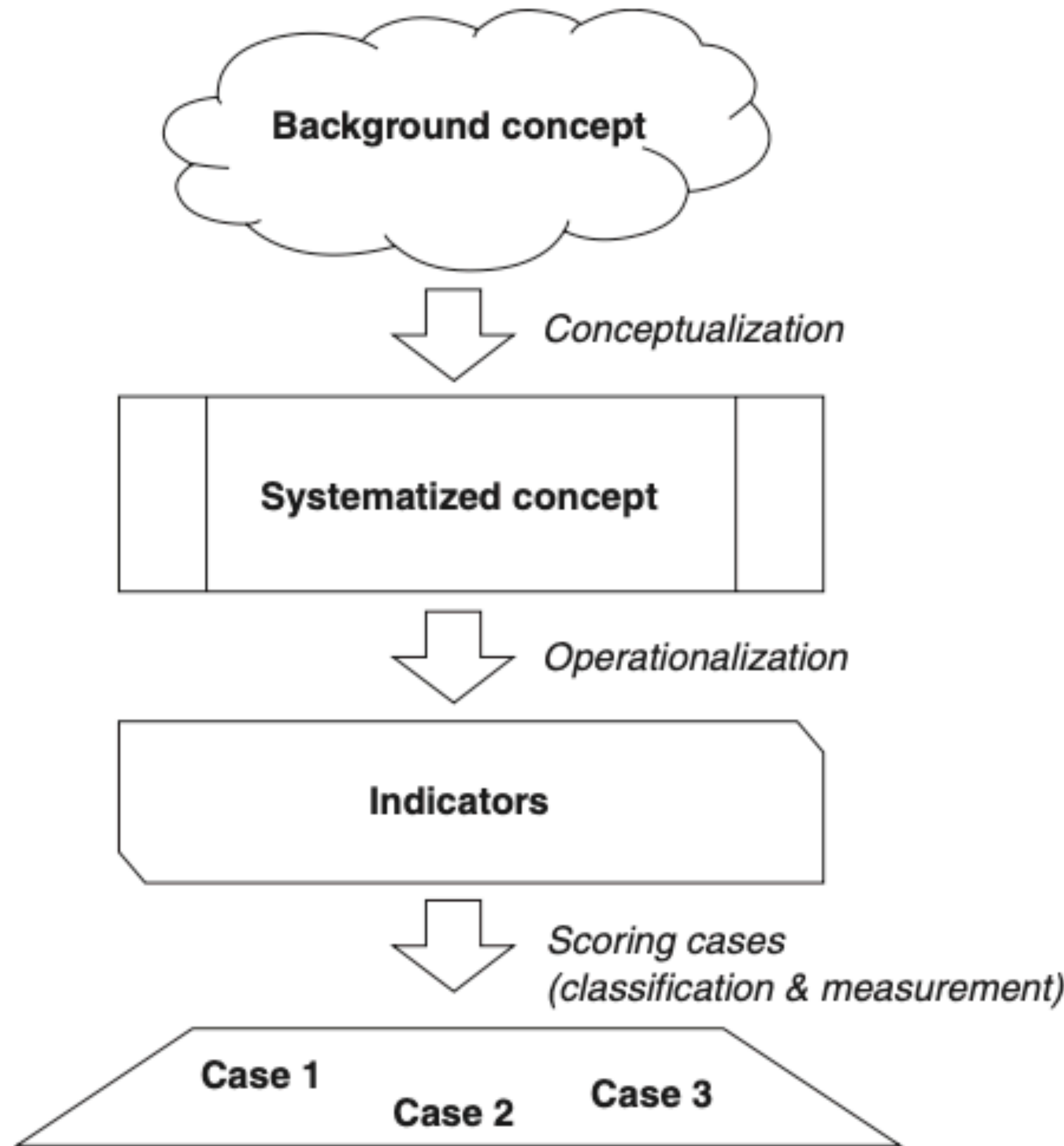


Measurement / Coding

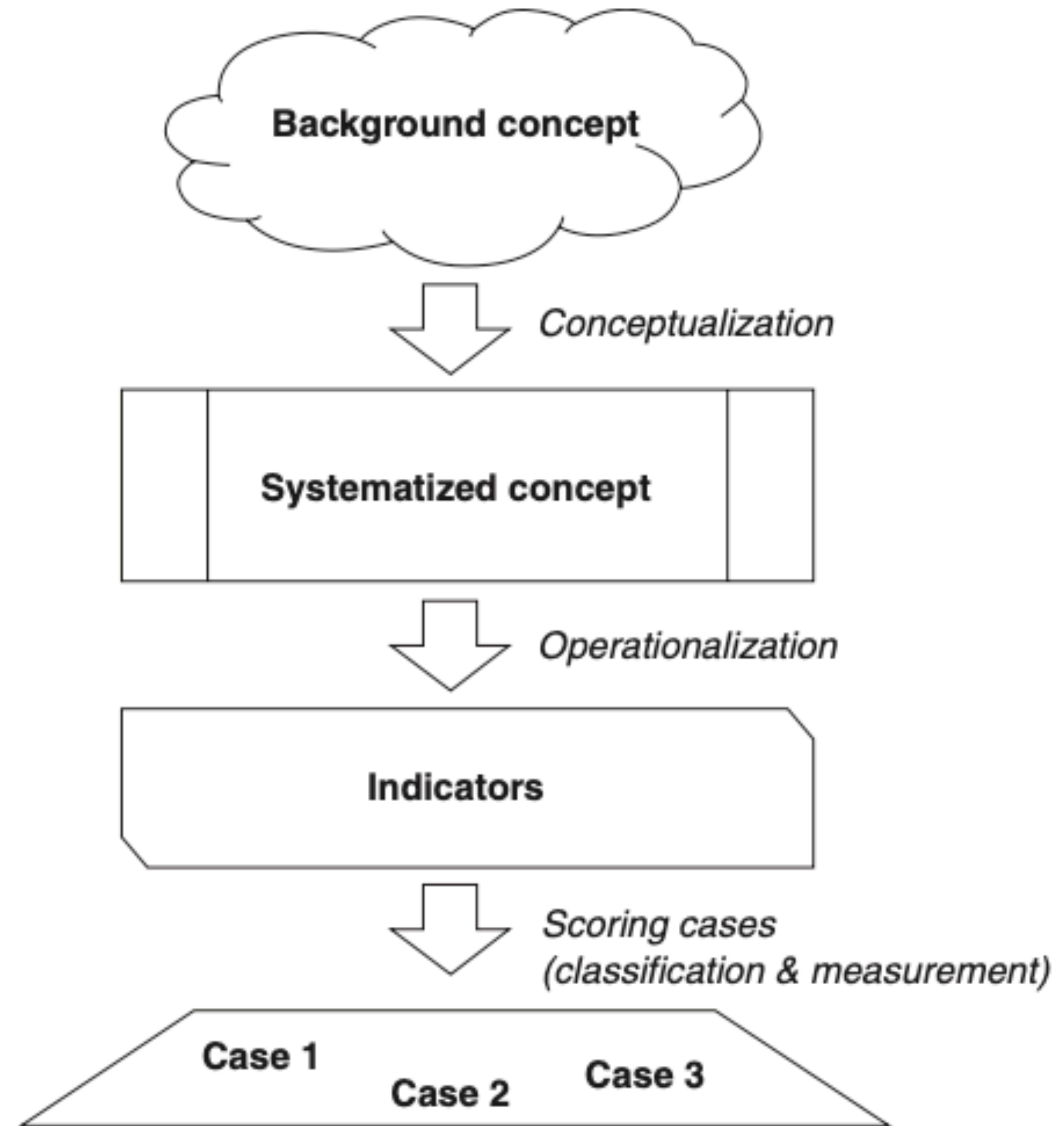
Once indicators are chosen, we must measure or code them:

- Surveys (e.g. Likert scales)
- Expert coding (e.g. ordinal ratings)
- Event counts / official data

The coding gives us numeric or categorical values for each indicator, which can then be aggregated.



- Democracy
- Contestation and Inclusion
- Regular election? Voting Right?
Effective party counts?
- Democracy Index



Sample V-Dem Coding / Survey Questions

Here are real items from the **V-Dem v15 codebook**:

- “To what extent are national elections free and fair?” (ordinal expert rating)
- “To what extent is there freedom of expression and alternative sources of media?”
- “To what extent do citizens have the right to associate in political parties or civil society organizations?”
- “To what degree do public policy decisions lie with elected officials rather than unelected actors?”

Now what?



A close-up photograph of a hand holding a black pen, marking a survey form. The form has five rows of checkboxes, each preceded by a label: 'Survey:', 'Excellent:', 'Good:', 'Fair:', and 'Poor:'. The first checkbox, corresponding to 'Survey:', is marked with a checkmark. The other checkboxes are empty. The hand is positioned at the top right of the form, and the pen is pointing at the first checkbox.

Survey:	☒
Excellent:	☐
Good:	☐
Fair:	☐
Poor:	☐

Aggregation & Index Construction (V-Dem)

After coding, V-Dem builds indices via:

1. Grouping indicators into subcomponents (e.g. election quality, civil liberties)
2. Using a latent variable / Bayesian measurement model to produce continuous estimates, accounting for coder differences, uncertainty, and bias

Aggregation & Index Construction (V-Dem)

After coding, V-Dem builds indices via:

1. Grouping indicators into subcomponents (e.g. election quality, civil liberties)
2. Using a latent variable / Bayesian measurement model to produce continuous estimates, accounting for coder differences, uncertainty, and bias
3. Combining subindices into high-level indices, like the Electoral Democracy Index (EDI / “v2x_polyarchy”)
4. Producing point estimates and uncertainty intervals for each country-year

Interpreting & Critiquing the Index?

The process we traced:

Background Concept (Democracy)

- Conceptualization — capture the meaning (Contestation and Inclusion)
- Operationalization — choosing Indicators (free elections, association, media freedom, etc.)

The process we traced:

Background Concept (Democracy)

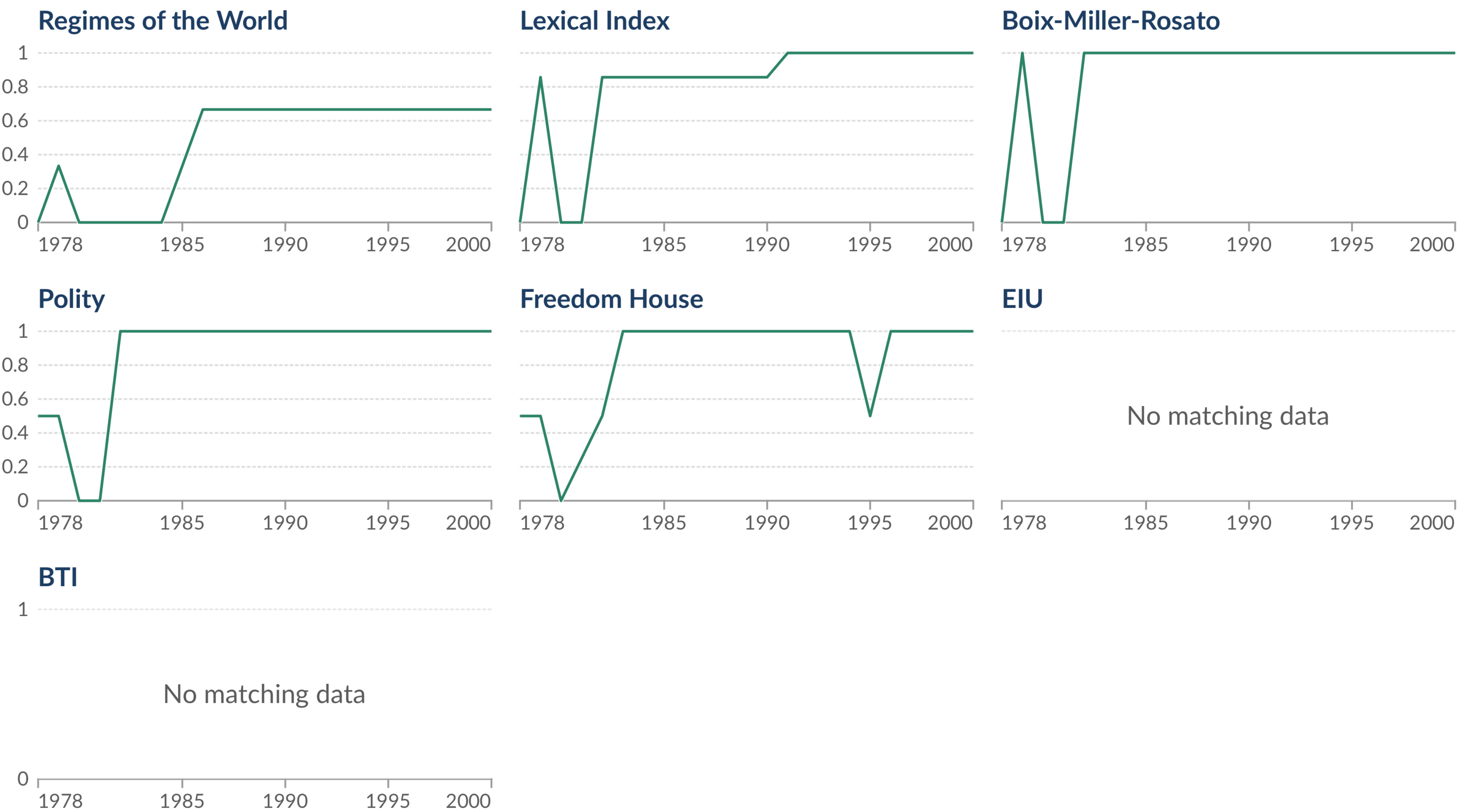
- Conceptualization — capture the meaning (Contestation and Inclusion)
- Operationalization — choosing Indicators (free elections, association, media freedom, etc.)
- Measurement — coding (expert ratings, ordinal scores)

Aggregation / Index (V-Dem EDI)

This example helps you see concretely how decisions at each step shape the final measure.

Political regime classification by source, Bolivia

The classifications are normalized to range from 0 to 1. The number of available sources differs by country or region.



POLI LIT Unit 1

Preview

POLI-LIT = 1 in-person lecture (today) + 3 online self-paced units

This Wednesday → Unit 1 opens online:

- Learn about the kinds of data used in political science
- What they look like and where they come from
- First steps before using data
- Basic statistical ideas (like “average” and “distribution”)

POLILIT Unit 1

Preview

Later in the semester (**Unit 2**):

- Intro to **R** (no download needed)
- Use **AI tools** to help with coding & data visualization

Each unit includes practice questions + open-ended reflection prompts

POLILIT Unit 1

Preview

Later in the semester (**Unit 2**):

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Each unit includes practice questions + open-ended reflection prompts

Complete the pre-course survey = earn 1 extra credit

Unit 1 = 5% of your course grade (completion-based) , ~ 1 hr

Realistic political data

- We created a mock dataset + codebook based on real sources
- You'll explore how data is used in research

Unit 1 covers:

- What political data looks like
- How to read a **codebook**
- What to check before you use data

Unit 1 (Oct 8)

Visible to students ▼

Introduction

Visible to students ▼

Understanding Data

Visible to students ▼

Practice: Reading Codebook

Visible to students ▼

Basic Statistical Concepts

Visible to students ▼

Ethical use of data

Visible to students ▼

Unit 1 Summary

Visible to students ▼

Unit 1 Feedback

Visible to students ▼

Due date: 10/10/25, 11:59 PM (EDT)

Basic statistical concepts:

- What is a mean, median, range?
- What does a “distribution” tell us?
- First steps in telling stories with data

Ethical thinking with data:

- How is data produced and collected?
- Who decides what gets measured and why?

Unit 1 (Oct 8)

Visible to students ▼

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